Persistence Mechanics of the Quantum Vacuum II: Operational Action and the Emergent Lagrangian

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Persistence Mechanics proposes an operational action that prices the maintenance of a system's required distinctions. We study conditional mathematical routes from this cost model to field dynamics on the E_8 substrate of Paper I. Lean verifies finite survival-weight concentration, phase-covariant transport, a first-order dispersion/Clifford equivalence, the sign conditions for a nonzero stable scalar minimum, and finite-dimensional reconstruction of unitary phase evolution under detailed balance and injectivity assumptions.

The concrete lattice results include the isotropic area tensor of the minimal D_4 triangle loops, a controlled Abelian constant-field continuum limit, two-sector chiral zero-mode conditions, and a detuning mass relation. These results distinguish exact lattice identities from continuum approximations. Further checked constructions connect actual glue-class reflection events to primitive cost and normalized trace scores. A specified phase-sensitive readout produces the cosine repair cost, and a unitary-eigenbasis link expansion has a controlled remainder. The selected fermion Hamiltonian has an exact signed spectrum and a local unitary implementation preserving its known vacuum modes. Physical carrier/representation selection, finite-tick spectral exhaustion, the microscopic readout and repair law, non-Abelian continuum response, physical normalization and emergent gravity remain to be established. Finite capacity alone does not prove exact elimination of higher-dimensional operators, a Born-rule derivation, or a unique Standard Model and Einstein–Hilbert action. The paper states the additional assumptions and remaining tests needed for those stronger conclusions.

I. INTRODUCTION

Persistence Mechanics (PM) develops the architecture, dynamics and operational action of persistent systems [1]. Its requirements are organized into Capacity, Map, Protocol, Governor, Toll and Margin. Paper I applies explicit lattice, readability, exchange and selection postulates to the closed vacuum and derives the E_8 substrate and its characteristic integers [2]. This paper applies that framework to the supplied vacuum substrate as a dynamical test: which update rules, cost functions and continuum responses can maintain the required distinctions? The general action framework is inherited; the task here is its physical realization, not a second derivation of PM.

The operational action is a proposed cost. Its exponential history weights favor lower-cost histories at positive inverse tolerance. A finite penalty suppresses a history without eliminating it; exact concentration requires a limit. Similarly, an exactly unitary lattice update and an approximate continuum equation are different objects. These distinctions determine the scope of the derivation.

A. Mathematical results and physical identifications

The Lean package checks several linked results. A first-order Hamiltonian has the stipulated relativistic squared dispersion exactly when its generators obey the corresponding Clifford relations. Local phase covariance requires a transporting link. A scalar potential quadratic in $s = |\phi|^2$ has a stable nonzero minimum under explicit no-starvation and no-runaway conditions precisely when its coefficients have the Mexican-hat signs. Finite survival weights admit a concentration limit; detailed balance and an injective symmetrized transfer matrix give a positive two-tick operator and unitary spectral phase evolution.

The new geometric and fermionic calculations make these questions concrete. The D_4 nearest-neighbour graph has triangles, whose summed area tensor is isotropic. The Abelian triangle action has a controlled constant-field continuum limit with a calculable normalization. Two D_4 sectors support a candidate Wilson/domain-wall mechanism with complementary chiral zero-mode conditions; separate results address walk twins, speed constraints and a detuning mass polynomial.

The new carrier/event calculation follows Paper I's actual minimal glue-class vectors and an integral E_8 root reflection into a counted primitive cost. A specified comparison readout then yields a trace score and the cosine phase cost, with an explicit primitive-event multiplicity. The corresponding link expansion now has a controlled finite-angle remainder. These results begin the common action/response bridge rather than completing its physical selection. In particular, the three-dimensional cubic walk and the D_4 gauge-loop geometry have not yet been proved to be one microscopic dynamics.

Each theorem has an explicit model boundary. The lattice geometry does not by itself select a physical gauge group, an occupation measure, a full update schedule or

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a numerical coupling. The two-sector construction is a candidate realization, not a uniqueness theorem derived solely from the sector count. Its exact spectrum, locality properties and response must be connected before numerical constants are inferred.

B. Structure and scope

Sections II and III state the cost model, survival interpretation and continuum assumptions. Sections IV to VI present the formalized fermion, gauge and scalar results. Section VII records the proposed gravitational interpretation and the premises still needed for it. Sections VIII and IX separate the proved constraints from the next completion tests. The appendices distinguish finite-dimensional quantum reconstruction from a measurement theory and record the uncompleted scattering and magnetic-moment calculations.

The aim is to derive physical laws and their constants from persistence constraints. The current results establish parts of that program, with assumptions exposed. They do not yet establish an exact, parameter-free Standard Model plus General Relativity or the exclusion of all other operators and particle content.

II. THE IMPORTED ACTION AND ITS APPLICATION

For this application we use a proposed specialization of the PM operational action [1]. It models the cost of maintaining required distinctions by a density

$$\mathcal{D} = \xi\,\mathbf{K}\,f, \qquad S_\Phi = \int \mathcal{D}\,d\mu, \tag{1}$$

where ξ is a specified primitive cost, $\mathbf{K}$ the maintained workload or complexity density, and f its operation rate. The state space, counted events, units and measure must be supplied for an application. Information capacity, description length and executed work are not interchangeable without that map. The multiplicative density is a modeling contract, not a proved consequence of statistical independence from coding or thermodynamic theorems.

A resource budget bounds the product and hence bounds f when ξ and $\mathbf{K}$ are positive (Lean: `PMAction.complexity_speed_bound`). It does not fix the factors individually. The economical selection principle is part of the application contract; this paper does not rederive a universal minimum-entropy-production theorem. A closed vacuum also cannot be treated as shedding entropy to an external bath without specifying the internal statistical description.

A. Least Action as Survivorship

For a finite history family with costs S_i, consider the supplied weights

$$P_\beta(i) = \frac{e^{-\beta S_i}}{\sum_j e^{-\beta S_j}}. \tag{2}$$

Lean proves positivity and normalization, multiplicative weights for additive costs, and the exact ratio

$$\frac{P_\beta(j)}{P_\beta(i)} = e^{-\beta(S_j - S_i)}. \tag{3}$$

For finite real costs and tolerance, every weight is strictly positive. Thus suppression is not exact deletion. For a fixed finite family, `Concentration` proves that the normalized distribution tends to the uniform distribution on its minimum-cost histories as $\beta \to +\infty$. A thermodynamic or long-observation application must justify its limit and effective inverse tolerance.

Stationarity alone is weaker than a minimum: `PMAction.stationary_not_minimal` supplies a counterexample. A nonnegative dissipative functional also cannot be equated directly with an unrestricted Lorentzian quadratic form (`lorentzian_cost_not_nonnegative`). The relation to physical time evolution therefore requires an additional construction, discussed next.

B. A Constructive Carrier–Event Cost Connection

The counting law can now be applied to explicit substrate carriers. `CarrierEvents` uses Paper I's minimal doubled vectors in each nontrivial D_4 glue class; every class has eight states, with no minimal vector omitted by the finite encoding. The reflection in $e_0 - e_1$ permutes each class reversibly and moves four of its eight states. `SubstrateReflection` connects that coordinate reflection to an integral E_8 root reflection preserving the actual quadratic form and glue labels, including its restriction through Paper I's $D_4 \oplus D_4$ embedding.

With explicitly uniform carrier sampling, a changed-state event has probability $1/2$ and cost $c_0/2$. More generally, a permutation P on $n > 0$ carrier labels has

$$p_{\text{changed}} = 1 - \frac{\operatorname{Re}\operatorname{Tr} P}{n}, \qquad C_{\text{changed}} = c_0 p_{\text{changed}} \tag{4}$$

(`CarrierCost`). The trace score here follows from counting fixed labels. This does not uniquely extend a permutation event to a continuous phase readout. Nor are eight distinct geometric vectors eight independent vectors: Lean verifies that the carrier vectors in four coordinates are linearly dependent. A complex amplitude space over their labels is an additional construction, not that cardinality identity.

a. Phase-sensitive comparison. For a specified unitary transport U and a reference branch, take the minus-port amplitude of a prepared basis label a to be $(e_a - Ue_a)/2$. Under the stated squared-amplitude readout, Lean derives its probability and the uniform mean

$$p_{\mathrm{flag}}(a) = \frac{1 - \mathrm{Re}\, U_{aa}}{2}, \qquad \overline{p}_{\mathrm{flag}} = \frac{1}{2}\left(1 - \frac{\mathrm{Re}\,\mathrm{Tr}\, U}{n}\right). \tag{5}$$

The two output probabilities are nonnegative and sum to one (`InterferenceReadout`). If a flag triggers m primitive repair events, weighted finite-event accounting gives $C = c_0 m\, \overline{p}_{\mathrm{flag}}$, without an independence assumption (`EventAccounting`, `CarrierResponse`). For the carrier reflection this flag probability is $1/4$, whereas its direct changed-state probability is $1/2$. The operational event must therefore be specified.

For a scalar phase $U = e^{i\theta}$ the repair cost is $(c_0 m/2)(1 - \cos\theta)$. A phase-independent required comparison cost remains a baseline B; subtracting the zero-phase reference gives that excess cost, rather than declaring comparison free. Reference preparation, uniform sampling, squared-amplitude readout and the physical repair multiplicity remain explicit inputs. This is not a derivation of the Born rule or a unique choice of those inputs from persistence.

III. FROM OPERATIONAL COST TO EFFECTIVE FIELD OPERATORS

The six PM pillars organize the candidate field terms. Their interpretation as a single physical action requires a common carrier state space, update rule, observable map and normalization. We distinguish the proved conditional structures from that remaining identification.

A. Measures, Costs and Evolution

1. Local densities

The local density must be distinguished from its spatial integral and from the accumulated history cost. For a specified spatial measure and time parameter, write

$$Z_\Phi(t) = \int_{\Sigma_t} \mathcal{D}(x,t)\, d\mu_t(x), \qquad S_\Phi = \int Z_\Phi(t)\, dt. \tag{6}$$

This is the spatial-then-temporal organization of the imported action, not an identification of a pointwise density with a system-wide rate. Combining the measures into a covariant spacetime integral requires the physical geometry and its volume element.

A continuum model may replace lattice counting by an integral and decompose a cost into local terms. In four dimensions a dimensionless action has a density of mass dimension four. This dimensional statement does not uniquely specify the terms, their coefficients, or a Lorentzian metric measure. Identifying $d\mu$ with $d^4x\sqrt{-g}$ is part of the proposed continuum gravitational description.

2. The statistical and unitary layers

Exponential survival weights define a statistical layer. In the finite-dimensional reconstruction, positive stationary weights and detailed balance make the symmetrized transfer matrix symmetric. Injectivity makes its square positive definite. Its eigenvalues define real logarithmic energies, and multiplication by $e^{-iE_i t}$ preserves the amplitude norm (`Reconstruction`). A deterministic bijection separately preserves that norm by permuting configuration amplitudes.

These are finite-dimensional results under named assumptions. They do not prove that detailed balance follows from zero total flux, that the statistical transfer matrix is injective, or that reconstructed time evolution is a nearest-neighbour update. They do not yet supply a quantum-field reconstruction, the Born rule or measurement dynamics; see appendix A.

B. The Candidate Additive Structure

We organize the proposed action as

$$\mathcal{L}_\Phi = \mathcal{L}_{\mathrm{CAP}} + \mathcal{L}_{\mathrm{MAP}} + \mathcal{L}_{\mathrm{PRO}} + \mathcal{L}_{\mathrm{GOV}} + \mathcal{L}_{\mathrm{TOL}} + \mathcal{L}_{\mathrm{MAR}}. \tag{7}$$

This bookkeeping does not prove statistical independence of six costs or identify six mutually orthogonal physical subspaces. Interactions and shared maintenance must be included without double counting. Costs incurred by coordination and savings relative to an uncoordinated baseline are distinct entries: a negative contribution to a net comparison is not a negative primitive expenditure. A Lorentzian Lagrangian need not be pointwise positive, so this accounting distinction does not make every displayed field term a positive dissipation density.

C. Kinematic Constraints

A nearest-neighbour update with spacing ℓ and step τ has a specified causal stencil. Choosing units $\ell/\tau = 1$ sets its link-speed scale; it does not by itself derive an isotropic relativistic dispersion, Lorentz transformations or equality of all field speeds. The walk calculations exhibit why this matters: the one-coin update has twins, while a particular twin-gapping split-step family reduces its light speed. The two-sector Hamiltonian is a different candidate. A shared low-energy metric requires matching these responses.

D. Dimensions and the Leading Operator List

For the stipulated first-order fermion and second-order boson kinetic terms, dimensional consistency gives

$$[\psi] = 3/2, \qquad [\phi] = [A] = 1. \tag{8}$$

Lean proves these relations and constraints on monomials of dimension at most four (`Operators`). In that list a fermion bilinear admits at most one additional unit of dimension, Yukawa and quartic scalar terms are allowed, and four-fermion and sixth-order scalar terms lie outside the cutoff.

The cutoff is a hypothesis defining the leading operator list. Finite state count alone does not prove it as an exact exclusion: `Potential.finite_states_do_not_force_quartic` distinguishes a sixth-order interaction on four amplitudes from every proposed quadratic-plus-quartic potential. With coefficients $c\ell^k$, higher-dimensional terms vanish as $\ell \to 0$ (`Operators.higher_dimension_suppressed`); they need not vanish at finite spacing.

1. *Mass terms and potentials*

The leading list permits a dimension-one fermion mass coefficient and scalar quadratic/quartic terms. Representation and gauge-covariance constraints must decide whether a bare mass or a scalar-mediated coupling is allowed. Orthogonality of two geometric sectors alone is not that representation-theoretic selection rule.

E. Target Continuum Form

The program seeks a response of the form

$$\mathcal{L}_\Phi = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi - \tfrac{1}{4}F^a_{\mu\nu}F^{a\mu\nu} - |D_\mu\phi|^2 - V(\phi) + \tfrac{M_P^2}{2}(R - 2\Lambda). \tag{9}$$

This is a target effective description, with consistent metric and field conventions required. The following sections prove parts of its operator structure for specified models. They do not yet derive its full representation content, gravitational dynamics or coefficients from the substrate.

IV. MAP AND CAPACITY: FERMION PROPAGATION

The fermion problem has three parts: construct an exactly norm-preserving update, control its low-momentum dispersion and species content, and identify any mass-generating coupling. The substrate's two D_4 sectors motivate the candidate below; they do not alone determine its Hamiltonian, representation or interaction charges.

A. Discrete Update versus Continuum Approximation

For Hermitian Hamiltonian matrices α_k, β, a central-difference approximation to the Dirac Hamiltonian is

$$H\psi = -i\sum_{k=1}^{3}\alpha_k\frac{\psi(x+e_k\ell) - \psi(x-e_k\ell)}{2\ell} + m\beta\psi. \tag{10}$$

Its forward-Euler step is

$$\psi(t+\tau) = \psi(t) - \frac{\tau}{2\ell}\sum_k \alpha_k[\psi(x+e_k\ell) - \psi(x-e_k\ell)] - i\tau m\beta\psi(t). \tag{11}$$

This is an approximation, not an exactly unitary lattice rule. Even the homogeneous mass-only Euler factor changes the norm at nonzero step (`DiscreteEvolution`). Formally dividing by τ gives

$$i\partial_t\psi = -i\sum_k \alpha_k\partial_k\psi + m\beta\psi. \tag{12}$$

The central-difference coefficient $\tau/(2\ell)$ is $1/2$ in link-speed units. It is a kinematic coefficient of this stencil, not a determination of an interaction coupling.

Exactly norm-preserving rotation-and-stream updates exist [3, 4]; `DiscreteEvolution` formalizes a finite-ring example. Its dispersion and twins must be checked separately. Likewise, exponentiating a self-adjoint Hamiltonian gives exact unitary evolution, but this does not prove a finite-range one-tick stencil: an exponential can propagate through repeated local hops. The domain-wall exponential remains conditional on that locality question.

B. The Clifford Constraint

For a first-order Hamiltonian $H(p,m) = \sum_k p_k\alpha_k + m\beta$, Lean proves the equivalence

$$H(p,m)^2 = (|p|^2 + m^2)\mathbb{I} \quad \text{for all } p, m \tag{13}$$

if and only if the generators square to one and pairwise anticommute (`Dirac.dispersion_iff_clifford`). This is a consequence of the specified dispersion, not of unitarity alone.

For the spacetime convention $\eta = \mathrm{diag}(-,+,+,+)$ one consistent choice is $\gamma^0 = \beta$, $\gamma^k = \beta\alpha_k$, with

$$\{\gamma^\mu, \gamma^\nu\} = -2\eta^{\mu\nu}\mathbb{I}. \tag{14}$$

Multiplying the Hamiltonian equation by β gives

$$(i\gamma^\mu\partial_\mu - m)\psi = 0. \tag{15}$$

The sign of the Clifford convention must not be mixed with a different metric convention. Deriving the physical representation and its continuum approximation remains distinct from the algebraic equivalence.

C. Twins and a Two-Sector Domain Wall

The one-coin streaming update obeys $W(k+\pi) = -W(k)$ (`Doubling.symbol_shift`). Its twin couples identically in every tested link background (`TwinTests.twin_anticommutes`). A particular two-coin partial-stream update gaps the twin, but its light speed is $|\cos r|$; restoring unit link speed ungaps the twin (`WalkSpeed`). These results rule out shortcuts within those specified update families.

A different candidate uses the two sectors as two slices of a Wilson/domain-wall Hamiltonian. This construction *does* use a Wilson term; no claim of a Wilson-free mechanism is made. At the chosen point $M = r = 1$, its zero-momentum inter-sector hops are $-\beta(1\pm\Gamma)/2$, where $\Gamma = i\beta\alpha_4$. Lean proves that the kernel conditions are

$$\Gamma x_1 = x_1, \qquad \Gamma x_2 = -x_2 \tag{16}$$

(`DomainWall.zero_modes`). The spatial momentum generators commute with Γ and their square gives $|k|^2$ under the stated Clifford hypotheses. This supports the proposed chiral leading response. The full squared operator and necessary energy branches are derived below. A concrete representation is now checked exactly in Lean:

$$\beta = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \qquad \Gamma = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \qquad \alpha_j = \begin{pmatrix} \sigma_j & 0 \\ 0 & -\sigma_j \end{pmatrix}, \tag{17}$$

where σ_j are the Pauli matrices. The module `ConcreteFermion` verifies the Clifford relations, chirality commutation and Hermiticity of the full eight-component Hamiltonian for real momenta and detuning. Each chirality projector has rank two. At $k = 0$, $\delta = 0$, the entire kernel is parametrized by

$$x_1 = (u_0, u_1, 0, 0), \qquad x_2 = (0, 0, u_2, u_3), \qquad u \in \mathbb{C}^4. \tag{18}$$

Lean proves this parametrization is a linear equivalence, hence the kernel has complex dimension four: two independent components on each wall, with opposite Γ chirality. This is a result about the specified Hamiltonian; persistence has not yet selected this representation or identified these components with physical species.

The script `calculations/domain_wall_check.py` checks a concrete matrix representation numerically. The corner mass polynomial has a proved positive bound (`DetuningMass.twin_gap`). The new `DomainWallSpectrum` result goes beyond corners: for the sine-momentum operator and Wilson detuning

$$s(k) = \sum_j \sin^2 k_j, \qquad \delta(k) = \sum_j (1 - \cos k_j), \tag{19}$$

a nonzero zero-energy vector anywhere in $[-\pi, \pi]^3$ requires $k = 0$. The proof uses the full matrix polynomial and does not assume the absence of off-corner zeros.

For every real-energy eigenpair it also proves the uniform bound $|E| < 8$ in the model's units. The sufficient step condition $0 < |\tau| \le 1/2$ then gives $|\tau E| < 2\pi$, so a unit eigenphase $e^{-i\tau E} = 1$ cannot occur away from $k = 0$. This supplies the scalar bandwidth premise on the entire momentum cell. It does not select the physical clock step, construct a finite-stencil exponential update, or assign the calculated zero-mode components to physical chiral gauge representations.

The interpretation of sectors as physical chiral gauge representations remains a separate task. Complementary zero-mode conditions do not prove that one sector is dynamically static or that a particular weak gauge group acts only on it.

D. An Exactly Local Unitary Split Update

The Hamiltonian calculation can be supplemented by a finite spatial stencil. Let S_j shift one coordinate on a finite periodic cubic lattice and put

$$A_j = \frac{\beta + i\alpha_j}{2}, \qquad \widehat{A}_j = \mathrm{diag}(A_j, A_j). \tag{20}$$

Lean checks $A_j^2 = 0$ and $A_j A_j^\dagger + A_j^\dagger A_j = I$. Consequently

$$B_j = S_j \otimes \widehat{A}_j + S_j^\dagger \otimes \widehat{A}_j^\dagger, \qquad B_j^\dagger = B_j, \quad B_j^2 = I, \tag{21}$$

and $R_j(\tau) = \cos\tau\, I + i \sin\tau\, B_j$ is exactly unitary and reads only the current site and its two neighbors along coordinate j (`LocalUnitary`, `SplitUpdate`). With $H_0 = H(0,3)$, define

$$U_\tau = (I_{\mathrm{space}} \otimes e^{-i\tau H_0}) R_0(\tau) R_1(\tau) R_2(\tau). \tag{22}$$

The on-site exponential mixes only eight internal components. Lean proves unitarity and conservation of the total squared amplitude on every finite periodic cubic lattice, together with a precise input-dependence theorem: one tick reads at most $3^3 = 27$ spatial sites, independently of lattice size (`PropagationBounds`, `WilsonSplit`).

The directional phase symbol is

$$B_j(k_j) = \mathrm{diag}(\cos k_j\, \beta - \sin k_j\, \alpha_j, \cos k_j\, \beta - \sin k_j\, \alpha_j). \tag{23}$$

The split generators satisfy $H_0 - \sum_j B_j(k_j) = H(k, \delta(k))$. The update's momentum symbol satisfies $U_0(k) = I$ and has first derivative $-iH(k,\delta(k))$ at $\tau = 0$ (`SplitGenerator`). This is a first-order consistency result. It does not establish equality with $e^{-i\tau H(k)}$ at nonzero tick size, a many-tick convergence bound, or the finite-tick zero-mode and eigenphase multiplicities. In particular the Hamiltonian bandwidth argument above cannot simply be transferred to this product. A separate double-precision diagnostic at $k = 0$, $\tau = 0.1$ reports a nearest eigenphase magnitude of approximately 7.47×10^{-4} rather than zero. This is a numerical warning, not a Lean obstruction theorem: the displayed ordering should not be treated as an exact vacuum-preserving physical update. This motivates separating the on-site factors as follows.

a. Vacuum-preserving ordering. Write $W = H(0,0)$ for the wall hopping alone, $\beta_8 = \mathrm{diag}(\beta, \beta)$ and

$$R_\beta(\tau) = \cos\tau\, I_8 + i \sin\tau\, \beta_8. \tag{24}$$

The corrected on-site coin and update are

$$\begin{aligned} \widetilde{Q}_\tau &= e^{-i\tau W}[R_\beta(\tau)^\dagger]^3, \\ \widetilde{U}_\tau &= (I_{\text{space}} \otimes \widetilde{Q}_\tau) R_0(\tau) R_1(\tau) R_2(\tau). \end{aligned} \tag{25}$$

At $k = 0$, each spatial factor has symbol $R_\beta(\tau)$, so Lean proves the exact identity

$$\widetilde{U}_\tau(0) = e^{-i\tau W} \qquad \text{for every real } \tau. \tag{26}$$

Since W annihilates the previously characterized four-dimensional vacuum kernel, every vector in that kernel is fixed exactly by $\widetilde{U}_\tau(0)$ (`VacuumSplit.vacuum_preserved`). The proof uses the vacuum projector and the exponential, with no small-tick approximation. The module `VacuumUniform` also proves this for the corresponding spatially constant states on every finite periodic lattice, connecting the symbol identity to the actual lattice update.

The corrected update is exactly unitary, conserves total squared amplitude, and has the same at-most-27-site spatial input bound. Separating the on-site factors changes only terms beyond first order: Lean checks $\widetilde{U}_0(k) = I$ and $\partial_\tau \widetilde{U}_\tau(k)|_{\tau=0} = -iH(k, \delta(k))$ (`VacuumGenerator`). Thus exact preservation of the known vacuum modes, unitarity, finite propagation and the stated Wilson tangent hold simultaneously in this construction.

This proves preservation of the known four modes; it does not prove they exhaust the fixed space of the finite-tick update. Additional unit eigenphases, including those away from the origin, a permissible physical tick, many-tick error control and physical species identification remain separate obligations. These constraints also do not establish that persistence uniquely selects this ordering.

E. Mass and the Scalar Bridge

Under the stated Clifford relations the hop products reduce to complementary projectors (`DetuningMass.hop_products`). Writing the diagonal operator as $K + \delta\beta$, Lean now proves the full squared matrix has diagonal blocks $(s + \delta^2)I + P_-$ and $(s + \delta^2)I + P_+$ and off-diagonal blocks $-\delta I$, with all momentum-hop cross terms cancelling (`DomainWallSpectrum.hamiltonian_square`). On the common $\Gamma = +1$ subspace this gives the reduced block

$$\begin{pmatrix} \delta^2 & -\delta \\ -\delta & \delta^2 + 1 \end{pmatrix}. \tag{27}$$

Lean checks that

$$\begin{aligned} m(\delta) &= \sqrt{\delta^2 + 1/4} - 1/2, \\ m(m+1) &= \delta^2, \qquad 0 \le m \le \delta^2. \end{aligned} \tag{28}$$

The full operator also obeys a polynomial implying, for every nonzero real-energy eigenvector,

$$E^2 = s + m(\delta)^2 \quad \text{or} \quad E^2 = s + (m(\delta) + 1)^2 \tag{29}$$

(`DomainWallSpectrum.energy_branches`). These are necessary spectral branches derived from the Hamiltonian, rather than roots postulated independently. For the concrete representation above, `ConcreteSpectrum.square_charpoly` strengthens this necessary condition to the exact identity

$$\det(\lambda I_8 - H^2) = [\lambda - s - m(\delta)^2]^4 [\lambda - s - (m(\delta) + 1)^2]^4. \tag{30}$$

The two roots are strictly separated, since $(m+1)^2 - m^2 = 2m + 1 > 0$. Lean proves each has algebraic multiplicity four (`ConcreteSpectrum.square_branch_multiplicities`). Thus both squared-energy branches occur in this model. The signed count is now established as well. A real matrix J with $J^2 = -I$ obeys $JH^T = -HJ$, giving equal characteristic polynomials for H and $-H$. Combining this pairing with the squared polynomial yields

$$\det(EI_8 - H) = [E^2 - s - m^2]^2 [E^2 - s - (m+1)^2]^2 \tag{31}$$

(`SignedSpectrum.signed_charpoly`). Whenever the light energy is nonzero, each of $\pm\sqrt{s + m^2}$ has algebraic multiplicity two; each of $\pm\sqrt{s + (m+1)^2}$ always has multiplicity two. At the origin the polynomial is $E^4(E-1)^2(E+1)^2$, consistent with the explicit four-dimensional zero kernel (`EnergyCounts`). These counts do not identify physical species. The detuning of each physical fermion is not yet derived; on-site detuning must not be identified with a change of the hop without an additional calculation.

A phase-covariant scalar link is a candidate interaction between sector states. Given representations and charges, its continuum operator can take the Yukawa form

$$\mathcal{L}_{\text{CAP}} = -y(\bar{\psi}_L \phi \psi_R + \bar{\psi}_R \phi^\dagger \psi_L). \tag{32}$$

The charge-pattern checks in Paper III are conditional on the chosen representations and scalar link. Identifying a background scalar with a linear physical mass $y\langle\phi\rangle$, and reconciling that law with the detuning spectrum, require a derived field and scale map. Neither a numerical Yukawa value nor its normalization follows from the hopping stencil alone.

The magnetic-moment response also remains to be calculated from the update; appendix B records the failure of the earlier winding-angle argument.

a. Comparison cost versus the fermion kinetic operator. `CommonResponse.wilson_form` rewrites the positive comparison penalty as $\mathrm{Re}\langle\psi, (2 - U - U^\dagger)\psi\rangle/4$. This is a transport-mismatch quadratic form. It does not alone derive the first-order Dirac action, fermionic statistics or the identification of glue-class labels with the spinor components used above. Those remain explicit carrier-to-response obligations.

V. PROTOCOL: PHASE-COVARIANT TRANSPORT AND GAUGE COST

Without a global phase reference, transport between independently rotated local frames must transform with those frames. `Protocol` proves that a fixed coefficient invariant under every independent phase rotation vanishes, whereas a transforming link preserves the transport expression. This supplies a conditional reason for a connection; it does not fix the gauge group or its coupling.

A. Links and Minimal Loops

For an Abelian connection write $U_\mu(x) = e^{-ig\ell A_\mu(x)}$. The observable link phase is unchanged by $g \mapsto g/c$, $A \mapsto cA$ for nonzero c (`Gauge.phase_normalization_freedom`). Thus a link-speed coefficient is not a determination of g.

For a supplied group, a product of links around a closed loop transforms by conjugation at its base point. Lean checks this for four links and shows that a conjugation-invariant score gives a gauge-invariant cost (`Gauge`). The choice of group, representation, loop set and score remains model data. The Wilson score is a candidate; invariance and positivity do not make it the unique possible class function.

The D_4 nearest-neighbour graph has 24 roots and admits triangles, so it cannot be treated as a square-only graph without extra structure. `D4Gauge.tri` contains 192 ordered root pairs with inner product one. Accounting for the six representations of an unoriented triangle across its three vertices gives a triangle density of 32 per site. The proved count is the ordered-pair count; this normalization must accompany its use in an action.

B. The Actual Cosine Cost and Its Continuum Approximation

For an Abelian loop phase, use $1 - \cos\theta$. Within the explicit comparison/repair contract of the preceding section, this functional form follows from the flag probability: W refreshes, each requiring m primitive repairs on a flag, cost $(c_0 mW/2)(1 - \cos\theta)$. Thus $\beta = c_0 mW/2$ within this event convention. The convention, refresh assignment and repair realization must still be justified for the physical substrate; symmetry alone does not select this comparison protocol. It is nonnegative and bounded by two (`Gauge.phase_cost_bounds`). It is not an exact quadratic function at finite phase; `wilson_not_exactly_quadratic` gives a counterexample. Finite node capacity does not cancel its higher Taylor terms.

The new `GaugeContinuum` module bounds the approximation. For spacing $h \neq 0$ and constant triangle coefficient a, define

$$Q_h(a) = \frac{1 - \cos(h^2 a)}{h^4}. \tag{33}$$

When $|h^2 a| \leq 1$, Lean proves

$$|Q_h(a) - a^2/2| \leq \frac{5}{96}|h|^4|a|^4, \qquad \lim_{h\to 0,\, h\neq 0} Q_h(a) = a^2/2. \tag{34}$$

This is a controlled small-spacing result. At fixed nonzero microscopic spacing it describes a long-wavelength approximation, not exact elimination of corrections.

C. The Geometric Quadratic Coefficient on D_4

For $B^p_{\mu\nu} = a_\mu b_\nu - a_\nu b_\mu$, the finite triangle sum satisfies

$$\sum_p B^p_{\mu\nu} B^p_{\rho\sigma} = 96(\delta_{\mu\rho}\delta_{\nu\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho}). \tag{35}$$

Thus for every antisymmetric field F,

$$\sum_p (F \cdot B^p)^2 = 192 \sum_{\mu,\nu} F^2_{\mu\nu} \tag{36}$$

(`D4Gauge.area_isotropic`, `flux_sum`). No added squares are needed for this isotropy. The determinant of the Cartan matrix is four; the cell-volume factor used in the coordinate realization is two.

For the supplied constant-field Abelian phase $h^2(g/4)(F \cdot B^p)$, sixfold triangle accounting and cell volume give

$$\mathcal{S}_h(F) = \frac{\beta}{12} \sum_p Q_h\big((g/4)(F \cdot B^p)\big). \tag{37}$$

Lean proves that the *actual cosine action* tends to

$$\lim_{h\to 0} \mathcal{S}_h(F) = \frac{\beta g^2}{2} \sum_{\mu,\nu} F^2_{\mu\nu}. \tag{38}$$

With $g \neq 0$ and $\beta = 1/(2g^2)$ this is the canonical Euclidean Maxwell coefficient $1/4$ (`GaugeContinuum.maxwell_limit`). The normalization identity does not select β or a physical gauge coupling.

This completion covers constant Abelian curvature. Nonconstant fields require control of geometric flux errors; non-Abelian fields additionally require path-order/BCH and representation-normalization bounds. A continuation from positive Euclidean cost to a physical Lorentzian theory is a further step, not simply the insertion of a minus sign. Scattering and vertex-response calculations remain open; see appendix B.

a. The event coefficient in the controlled limit. `CarrierResponse.common_response_limit` substitutes the counted coefficient into the already proved triangle limit, giving

$$\lim_{h\to 0} \mathcal{S}_h(F) = \frac{c_0 m W g^2}{4} \sum_{\mu,\nu} F_{\mu\nu}^2. \tag{39}$$

This ties the coefficient to a primitive toll, repair multiplicity and refresh count in one specified contract. Canonical coefficient $1/4$ requires $c_0 m W g^2 = 1$ in these conventions. For matrix-valued holonomies, representation traces and indices must also be included; this Abelian result does not determine the physical relative gauge normalization or which resource each gauge sector refreshes.

VI. GOVERNOR: STABLE NONZERO SCALAR OCCUPANCY

Assume a scalar whose gauge-invariant radial variable is $s = |\phi|^2 \geq 0$, and restrict the leading potential to $V(s) = as + bs^2$ with $V(0) = 0$. Phase invariance permits functions of s; it does not by itself select a polynomial of degree two. The dimension-four restriction defines the leading ansatz rather than an exact consequence of finite state count.

A. The Sign Selection Theorem

`Governor.governor_selects_hat` proves an equivalence. Some positive occupancy costs less than the empty state, and the potential attains a minimum at positive occupancy, precisely when $a < 0 < b$. Writing $a = -\mu^2$, $b = \lambda$ gives

$$V(\phi) = -\mu^2|\phi|^2 + \lambda|\phi|^4, \qquad \mu^2 > 0, \quad \lambda > 0. \tag{40}$$

The no-starvation and no-runaway conditions are explicit selection hypotheses. This theorem fixes the signs within the ansatz, not the numerical coefficients or the physical identity of the scalar.

B. The Minimum and Field Convention

Completing the square gives

$$V(s) + \frac{\mu^4}{4\lambda} = \lambda\left(s - \frac{\mu^2}{2\lambda}\right)^2. \tag{41}$$

Lean proves the lower bound and its equality condition (`Potential`). With the convention $\langle|\phi|^2\rangle = v^2/2$, the minimum condition is $\mu^2 = \lambda v^2$. This avoids confusing v with the radial amplitude $v/\sqrt{2}$.

The candidate kinetic term is $-|D_\mu\phi|^2$ in the stated metric convention. Its coefficient, representation and covariant derivative must be derived from the same microscopic response used for the fermion and gauge sectors. Canonical normalization and a physical scalar mass do not follow from the potential minimum alone. Paper III tests proposed coefficients, including their joint mass relations; they are not assumed uniquely fixed here.

C. Finite Spacing and the Ultraviolet Question

Finite lattice resolution can regulate a specified dynamics, but a regulator alone does not prove naturalness, the observed hierarchy, or exact absence of higher operators. The Lean finite-amplitude counterexample retains an independent sixth-order interaction. Higher-dimensional operators with coefficients $c\ell^k$ vanish in the stated scaling limit; their finite-spacing effects require bounds.

Deriving the scalar kinetic response, its allowed higher terms and its relation to the detuning spectrum is required before claiming a physical Higgs sector or a resolution of the hierarchy problem. The Planck/Higgs ratio is not proved in this paper.

D. The Link Response and Its Remaining Normalization

The same specified comparison functional extends to transported amplitudes as $C_{\text{link}} = (c_0 m/4)\|\phi' - U\phi\|^2$ (`CommonResponse`). Sharing the prefactor is conditional on sharing the readout and primitive repair multiplicity; this algebra does not prove that every physical sector realizes the same repair process. Unnormalized field amplitudes also require a map to maintained workload, rather than treating an arbitrary squared norm as a probability for one event.

For a Hermitian generator in a unitary eigenbasis, $\Theta = V\,\mathrm{diag}(\lambda)V^\dagger$, let $U_\epsilon = V\,\mathrm{diag}(e^{i\epsilon\lambda})V^\dagger$ and $c = V^\dagger\phi$. When $|\epsilon\lambda_k| \leq 1$, `LinkExpansion` proves the controlled bound

$$\left|\|\phi - U_\epsilon\phi\|^2 - \epsilon^2\|\Theta\phi\|^2\right| \leq \frac{5}{48}\epsilon^4 \sum_k |c_k|^2\lambda_k^4. \tag{42}$$

Thus the leading link penalty used for the scalar/gauge mass response has an actual finite-angle error bound. Selecting its physical representation, scalar kinetic normalization, field-to-workload map and fermion coupling still requires the common microscopic carrier dynamics.

VII. TOLL AND MARGIN: THE GRAVITATIONAL PROPOSAL

The proposed interpretation associates metric response with maintaining causal connections (Toll), and a volume

term with a baseline resolution cost (Margin). The current Lean package does not derive metric dynamics, an Einstein equation, a spin-two spectrum, or a cosmological constant from the substrate. This section states the target and the additional work it requires.

A. Capacity and Metric Response

A proposed available-capacity ledger is

$$C_{\text{available}}(x) = \nu - \mathbf{K}(x). \tag{43}$$

This accounting identity does not specify a metric. In particular, the suggested weak-field relations

$$g_{00} \simeq -\left(1 - \frac{\mathbf{K}}{\nu}\right), \qquad \Phi \simeq -\frac{\mathbf{K}}{2\nu} \tag{44}$$

require a derived propagation/response law and the identification $g_{00} \simeq -(1 + 2\Phi)$. Neither follows from the deficit alone. A refractive-index interpretation also requires its own transport equation.

Using the same symbol $\mathbf{K}$ in inertial and gravitational ledgers is not a proof of equal physical masses. A completion must compute both responses from the same maintained state and establish their normalization and coupling to probes.

B. The Einstein–Hilbert Target

The candidate continuum action is

$$S_{\text{gravity}} = \int d^4x\sqrt{-g}\,\frac{M_P^2}{2}(R - 2\Lambda). \tag{45}$$

Uniqueness results for a four-dimensional, local, coordinate-invariant metric theory with the specified second-order field-equation assumptions [5] motivate this target. Those assumptions must first be obtained from the lattice. A nearest-neighbour microscopic stencil alone does not prove exact exclusion of higher continuum derivatives or curvature terms.

A proof would need an explicit geometric state and update, a metric observable, a controlled continuum limit, its symmetry identities and propagating-mode spectrum. No ghost-free or pure spin-two theorem is presently supplied, even at linear order. The bare stiffness, the Einstein–Hilbert normalization and a physical equivalence-principle statement remain open.

C. The Volume Term

Identifying a baseline resolution cost with $\rho_{\text{vac}} = M_P^2\Lambda$ is a proposed matching condition. The Margin pillar does not yet prove that the effective vacuum contribution is static or that all additional scalar dynamics are forbidden. Consequently no exclusion of dynamical dark energy is inferred here. A completed model must derive the volume response and its conservation law alongside the metric dynamics.

VIII. SYNTHESIS AND PROOF BOUNDARIES

The checked results provide several components of a possible persistence-based dynamics: finite survival selection, phase-covariant transport, dispersion/Clifford equivalence, geometric gauge isotropy, conditional unitary reconstruction, scalar sign selection and a candidate two-sector fermion mechanism. They are not yet a derivation of a unique complete physical action.

A. Leading Operators versus Exact Exhaustion

Within an assumed dimension-four list, `Operators.allowed_monomials` bounds the number of fields and derivatives. This classification does not prove the physical cutoff itself, determine representations, or exclude all other invariants. The actual cosine action and the finite-amplitude scalar counterexample demonstrate why higher terms cannot be deleted solely by invoking finite capacity.

The controlled D_4 gauge limit is a positive replacement for exact truncation: the leading Abelian constant-field response follows with a bound on each phase's error. Nonconstant and non-Abelian response, fermion continuum control and gravitational dynamics need analogous proofs.

B. Matter and Radiation

The product $\xi\mathbf{K}f$ motivates a complexity/rate tradeoff under a fixed budget. By itself it does not classify physical excitations into exactly two stable phases. A classification of massive and massless modes requires the actual update's spectrum, conserved observables and stability conditions. The candidate domain-wall construction is one step toward such a spectrum; it does not prove the proposed universal matter/radiation bifurcation.

C. Connection to Numerical Constants

A field normalization, scale convention and common response must be fixed before operator coefficients are identified with measured constants. Paper III tests its proposed formulas jointly. Improving a storage-overhead mechanism cannot substitute for closing the scalar, fermion and gauge response relations at a common normalization.

IX. COMPLETION TESTS AND FALSIFIABILITY

A falsifiable prediction requires a specified physical model. The current formal results identify concrete tests of candidate updates rather than a completed exclusion of all alternative particle content or gravity theories.

1. **Fermion spectrum.** Derive the full momentum-dependent two-sector spectrum, multiplicities and chirality. Prove that no unwanted zeros remain away from the origin, rather than checking only corners or a numerical grid.

2. **Discrete time and locality.** Construct an exactly unitary update with the required causal stencil, or bound the propagation tails of an exponential update. Prove a bandwidth condition that excludes phase aliasing across the entire spectrum.

3. **Shared propagation response.** Derive common low-energy light cones and controlled finite-spacing deviations for fermion, gauge and scalar sectors. Unit link speed alone is insufficient.

4. **Gauge and scalar response.** Extend the constant Abelian triangle result to nonconstant and non-Abelian fields. Derive kinetic normalizations, representations and coupling coefficients. Test the scalar/fermion mass map rather than equating detuning with a linear Yukawa mass by definition.

5. **Dynamical observables.** Compute scattering, vertex response and vacuum polarization from the completed update and its species content. The current cross-section comparison and assigned Schwinger factor are not those calculations.

6. **Gravity.** Construct the metric response, its continuum equations and physical spectrum. Only then test a stiffness normalization, equivalence principle or cosmological prediction.

The present proofs do not exclude a fourth generation, supersymmetry, higher spins, higher-dimensional effective interactions or evolving dark energy. Such exclusions would require representation, dynamics and observable theorems not currently established. If a completed model predicts them, its assumptions and quantitative tests should be stated at that stage.

X. CONCLUSION

Persistence Mechanics motivates an economical description of the work needed to maintain a system's required distinctions. For explicit mathematical models, Lean now verifies survival-weight concentration, phase-covariant transport, a dispersion/Clifford equivalence, scalar sign selection and finite-dimensional routes to norm-preserving evolution. These results expose their state, symmetry, cost and limiting assumptions.

The concrete advances now include the D_4 triangle area tensor and controlled Abelian constant-field limit, actual glue-class carrier/reflection event counts, conditional phase-sensitive trace/cosine costs and a controlled link expansion. The specified Wilson/domain-wall Hamiltonian has a checked signed spectrum; its local unitary completion preserves the known vacuum modes. Exclusion of additional finite-tick modes remains open. The central priority is to connect the actual carrier dynamics, primitive event law and fermion/gauge/scalar responses in one model, then extend its continuum estimates.

The physical carrier identification, common response normalization, gauge representations and numerical coefficients remain to be derived. The gravitational action and the measurement interpretation remain proposals. Finite capacity is not treated as a proof of exact operator exhaustion or a resolution of the hierarchy problem. Paper III's constants must be tested against the same response once it is constructed; agreement of separately proposed formulas does not establish joint closure.

The program therefore has a precise next step: connect the lattice update, its spectrum and its observable response in one model. That would turn the current operator-level and geometric results into stronger physical claims, with explicit conditions for their validation or failure.

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Appendix A: Quantum Reconstruction and the Remaining Measurement Problem

The formal results separate deterministic evolution from statistical history selection. A bijection of a finite configuration space permutes arbitrary complex amplitudes and preserves their squared norm (`Reconstruction.bijective_update_unitary`). This construction does not select amplitudes or a measurement rule.

For a statistical transfer matrix T and positive weights π, detailed balance $\pi_x T_{xy} = \pi_y T_{yx}$ makes

$$S_{xy} = \sqrt{\pi_x}\, T_{xy} / \sqrt{\pi_y} \tag{A1}$$

symmetric. If S is also injective, S^2 is positive definite. Its positive eigenvalues define real logarithmic energies, and spectral phase evolution $c_i(t) = e^{-iE_i t} c_i(0)$ preserves $\sum_i |c_i|^2$. Lean verifies these finite-dimensional steps. Detailed balance, injectivity and the relation between this transfer model and the microscopic update are premises. A full quantum-field reconstruction, causal locality and a measurement interpretation do not follow from the spectral calculation alone.

1. Survival Weights Are Not Yet a Born-Rule Derivation

For nonzero amplitudes, assigning $S_k = -\log |c_k|^2 + S_{\text{base}}$ gives weights proportional to $|c_k|^{2\beta}$. At $\beta = 1$ their normalization reproduces $|c_k|^2$. This is an algebraic conditional statement: it puts the squared amplitudes into the assigned action and does not derive that assignment or the value of β. No Born-rule theorem is currently claimed from this argument.

2. Correlations and Projection

The orthogonal-sector decomposition does not, by itself, construct an entangled state, Bell correlations or a no-signalling measurement theory. Equal internal coordinates do not remove a nonzero spacetime contribution to an orthogonal distance formula. An instantaneous internal update is therefore not inferred merely from a shared internal label. A physical account requires a joint state space, local observables and their causal evolution.

3. Measurement and Basis Selection

A projection rule $P_k\psi / \sqrt{\langle \psi, P_k \psi \rangle}$ can be stated once an observable, probability law and apparatus interaction are supplied. They are not selected by the current node-capacity count. No proof presently connects measurement to a unique capacity-exhaustion threshold, an entropy increment $\log 2$, or a preferred basis. These remain questions for a completed microscopic interaction model, rather than conclusions of the finite reconstruction theorem.

Appendix B: Dynamical Validation of the Emergent Lagrangian

This appendix records proposed response identifications and their present limits. The discrete-to-continuum operator results do not yet provide a scattering or vertex calculation. Boundary matching and the physical coupling normalization must be derived from the same update before these comparisons become predictions.

1. The Admittance Amplitude ($\sqrt{\alpha}$)

A localized topological defect maintains stability against the baseline vacuum impedance ($Z_{\text{vac}} = \alpha^{-1}$) by operating as a resonant cavity. In this representation, an interaction constitutes a **Topological Splice**—a geometric operation that conserves local information density while allowing the defect to shed accumulated stress via a gauge coordination field (the candidate gauge field of section V).

Because the transition probability of the operational action scales with the inverse of the substrate's geometric impedance, the probability amplitude ($\mathcal{A}$) of this coupling event is defined by the coordinate admittance of the lattice node:

$$\mathcal{A} \propto \frac{1}{\sqrt{Z_{\text{vac}}}} = \sqrt{\alpha} \tag{B1}$$

The identification $e = \sqrt{4\pi\alpha}$ is a proposed physical normalization. The square-root amplitude convention and its relation to a substrate transition probability remain to be derived; the link phase alone does not fix the field normalization.

2. Dynamic Validation: Scattering Cross-Section

For comparison with electron-positron annihilation ($e^+e^- \to \mu^+\mu^-$), the heuristic construction combines the following assigned factors. A discrete scattering calculation has not yet derived them:

1. **Coupling Efficiency (η_C):** The total transition probability scales with the joint product of the input and output admittances. Applying the vacuum admittance ($\sqrt{\alpha}$) established in the substrate architecture yields:

$$\eta_C = (\sqrt{\alpha} \cdot \sqrt{\alpha})^2 = \alpha^2 \tag{B2}$$

2. **Geometric Projection (1/3):** The intermediate vector field propagates through the **Interaction Remainder** ($\sigma - \chi = 3$), which represents the unconstrained spatial dimensions of the manifold bulk. Projecting this isotropic 3D bulk flux onto the 1D propagation axis introduces a geometric normalization factor of:

$$\frac{1}{\sigma - \chi} = \frac{1}{3} \tag{B3}$$

3. **Causal Scaling ($1/s$):** With dimensionless couplings and no additional scale in the proposed high-energy approximation, the comparison uses inverse center-of-mass energy squared (s):

$$\text{Scale} \propto \frac{1}{s} \tag{B4}$$

Integrating these three structural factors with the circular boundary of the projected interaction plane (4π) yields the high-energy cross-section:

$$\sigma_{E8} = \frac{4\pi\alpha^2}{3s} \tag{B5}$$

This matches the high-energy limit of standard quantum electrodynamics. The match is a consistency check rather than a derivation: the projection factor 1/3 and the boundary factor 4π are identified with substrate quantities, but they are not yet produced by the discrete update rule. A derivation would compute the cross-section from the update and its gauge coupling.

3. The Schwinger Limit: Status

The first-order anomalous magnetic moment of the electron is

$$a_e = \frac{\alpha}{2\pi} \approx 0.0011614. \tag{B6}$$

This is the textbook one-loop QED result, and it is the target here. The geometric reading identifies it as the coupling flux per gauge cycle, $a_e = \alpha/2\pi$, with the 2π supplied by the single-cover boundary loop. That reading reproduces the correct value, but as a heuristic: the factor $1/2\pi$ is assigned rather than derived from the update rule. A double-cover winding ratio of the form $(4\pi/2\pi) \cdot (g^2/4\pi)$ does not lead to it: with $g^2/4\pi = \alpha$ it equals 2α (Lean: `AppendixChecks.stated_product`, `not_schwinger`).

Obtaining a_e from the substrate remains open. A derivation must compute the vertex response of the discrete Dirac update in a background field, rather than assemble the result from a ratio of winding angles. The same calculation is needed for the vacuum-polarization coefficient that sets the running of α; the discrete update's species content, including its twins (`Doubling`, `TwinTests`), must be fixed first.